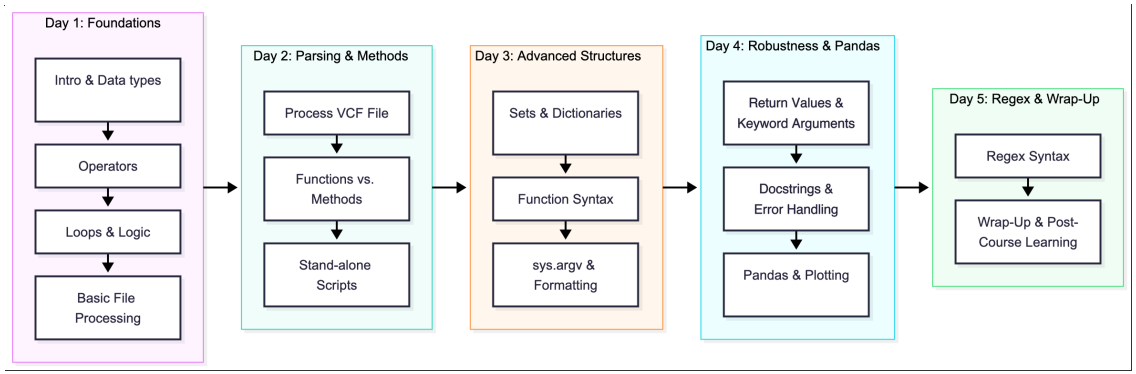
Introduction to



with Application to Bioinformatics

- Day 5

Yesterday's Key Takeaways



- Error handling using `try..except`, which helps you catch problems and prevent your program from crashing.
- Keyword arguments and positional arguments, which give you more control, readability, and flexibility when calling functions.
- Docstrings and comments, which provide clear documentation and make your code more understandable and maintainable.
- Pandas, a powerful and widely used library for working with tabular data.

Day 5

- **Session 1**
    - Quiz: Review of Day 4
    - Lecture: Go through questions
    - Lecture: Introduction to regex
    - Ex1: Find the pattern using regex
- **Session 2**
    - Lecture: Regex in Python
    - Ex2: Regex using Python
    - PyQuiz 5.1
- **Session 3**
    - Lecture: Sum up
    - Presentation about NBIS
    - Ex3: Final exercise
- **Project time**

Quiz: Review Day 4

Go to Canvas, `Modules -> Day 5 -> Review Day 4`

~15 minutes

1. What happens if you declare a variable with the same name inside and outside a function?

The variable inside the function has a separate scope and does not affect the one outside

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The variable inside the function has a separate scope and does not affect the one outside

In [ ]:
```
name = "Max"
def changeName():
    name = "Niko"
    print(f"name inside the function: {name}")
changeName()
print(f"name outside of the function: {name}")
```

2. What is the difference between positional arguments and keyword arguments?

Keyword arguments can be given in any order, while positional arguments depend on the function's order

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Keyword arguments can be given in any order, while positional arguments depend on the function's order

In [ ]:
```
def sum3(a, b, c):
    print(f"a={a}")
    print(f"b={b}")
    print(f"c={c}")
    return a+b+c
sum3(1, c=2, b=3)
```

2. What is the difference between positional arguments and keyword arguments?

Keyword arguments can be given in any order, while positional arguments depend on the function's order

In [ ]:
```
def sum3(a, b, c):
    print(f"a={a}")
    print(f"b={b}")
    print(f"c={c}")
    return a+b+c
sum3(1, c=2, b=3)
```

Arguments can be used in both ways, with or without keyword, if there is no ambiguity

- When used with keyword, they are keyword arguments
- When used without keyword, they are positional arguments

3. What will be the output of the following code snippet?

In [ ]:
```
def add(x=0, y=1, z=0):
    return x + y + z
print(add(1, 2))
print(add(1, y=2, z=3))
print(add())
```

4. Why is it beneficial to use docstrings in functions?

- They provide explanations and details about the function for others reading your code. **Both `"""` and `'''` can be used for docstring**

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- They provide explanations and details about the function for others reading your code. **Both `"""` and `'''` can be used for docstring**

In [ ]:
```
def add(x, y, z=0):
    """This function adds up three numbers, with the last one default to
    # Here we secretly increment x by 1
    x += 1
    return x + y + z
help(add)
```

5. How can you see the documentation of a Python library function in the console?

- Use help(library.function)

In [ ]:
```
help(print)
```

6. Which of these import statements would avoid a name conflict if there's a local variable `math` in the same script?

- import math as m

In [ ]:
```
import math as m
math = "great"
m.sqrt(5)
```

7. What will happen if you import the same module multiple times in a Python script?

- Python ignores subsequent imports of the same module in the same script

If you run

```
import myMoudle
```

and then update `myMoudle` and then reload with `import myMoudle` in Jupyter notebook, the module will not be updated.

In [ ]:
```
import mylib
print(mylib.add2(5,8))
```

7. What will happen if you import the same module multiple times in a Python script?

- Python ignores subsequent imports of the same module in the same script

If you run

```
import myMoudle
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and then update `myMoudle` and then reload with `import myMoudle` in Jupyter notebook, the module will not be updated.

In [ ]:
```
import mylib
print(mylib.add2(5,8))
```

In [ ]:
```
# Update mylib.py, add a new function multi2
print(mylib.multi2(5,8))
```

7. What will happen if you import the same module multiple times in a Python script?

- Python ignores subsequent imports of the same module in the same script

If you run

```
import myMoudle
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and then update `myMoudle` and then reload with `import myMoudle` in Jupyter notebook, the module will not be updated.

In [ ]:
```
import mylib
print(mylib.add2(5,8))
```

In [ ]:
```
# Update mylib.py, add a new function multi2
print(mylib.multi2(5,8))
```

In [ ]:
```
if 'mylib' in sys.modules:
    del sys.modules['mylib']
import mylib
print(mylib.multi2(5, 8))
```

8. If you want to filter rows in `df` where `age` is greater than 30, which command would you use?

- df[df['age'] > 30]

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- df[df['age'] > 30]

In [ ]:
```
import pandas as pd
df = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie', 'David'],
    'age': [25, 30, 35, 40],
    'height': [165.4, 175.3, 168.5, 180.6]
})
print(df)
df[df['age'] > 30]
```

9. If you want to rename multiple columns in a DataFrame `df`, which method should you use?

- df.rename(columns={'old_col1': 'new_col1', 'old_col2': 'new_col2'})

If you don't specify the key `columns`, it renames the rows

In [ ]:
```
import pandas as pd
df = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie', 'David'],
    'age': [25, 30, 35, 40],
    'height': [165.4, 175.3, 168.5, 180.6]
})
df.rename(columns={'name': 'SuperStar', 'age': 'TooOld'}, index={0:15, 1
```

Regular Expressions

- A regular expression (regex or regexp) is a sequence of characters that defines a search pattern.
- Use case: regular expressions are used in text processing for searching, matching, and manipulating strings

In [ ]:
```
!ls *.py
```

Examples where regex can play a role

- Find variations in a protein or DNA sequence
    - `"MVR???A"`
    - `"ATG???TAG"`
- American/British spelling, endings and other variants:
    - salpeter, salpetre, saltpeter, nitre, niter or KNO3
    - hemaglobin, heamoglobin, hemaglobins, heamoglobin's
    - catalyze, catalyse, catalyzed...
- A pattern in a VCF or GTF file
    - a digit appearing after a tab

Regex is not unique for Python and it is supported by

- most programming languages,
- text editors
- command line tools (e.g. `grep`, `sed`)
- search engines

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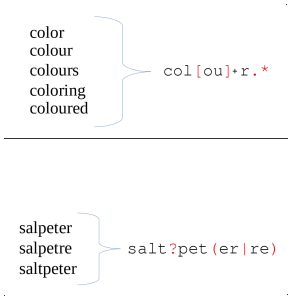
- most programming languages,
- text editors
- command line tools (e.g. `grep`, `sed`)
- search engines

In [ ]:
```
!grep "furniture.*sell" ../downloads/blocket_listings.txt
```

A playground for regex with detailed explanations of your regex

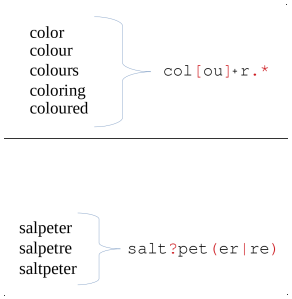
https://regex101.com/

Defining a search pattern



- `.` matches any character (once)
- `?` repeat previous pattern 0 or 1 times
- `*` repeat previous pattern 0 or more times
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In [ ]:
```
colou?r.*
```

More common operations - classes of characters

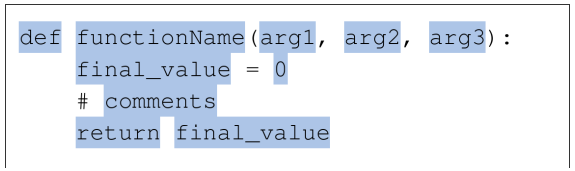
- `\w` matches any letter or number, and the underscore
- `\d` matches any digit
- `\D` matches any non-digit
- `\s` matches any whitespace (spaces, tabs, ...)
- `\S` matches any non-whitespace

More common operations - classes of characters

- `\w` matches any letter or number, and the underscore

`\w+`

```
def functionName(arg1, arg2, arg3):
    final_value = 0
    # comments
    return final_value
```



More common operations - classes of characters

- `\d` matches any digit

`\d+`

```
def functionName(arg1, arg2, arg3):
    final_value = 0
    # comments
    return final_value
```

More common operations - classes of characters

- `\s` matches any whitespace (spaces, tabs, ...)

`\s+`

```
def functionName(arg1, arg2, arg3):
    final_value = 0
    # comments
    return final_value
```

More common operations - classes of characters

- `[abc]` matches a single character defined in this set {a, b, c}
- `[^abc]` matches a single character that is **not** a, b or c
- `[a-z]` matches all letters between `a` and `z` (the english alphabet).
- `[a-z]+` matches any (lowercased) english word.

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- `[a-z]+` matches any (lowercased) english word.

```
plaintext
Three cats and 11 dogs run fast across grass fields.
```

In [ ]:
```
# matches any single character not a letter and not a space.
[^a-zA-Z ]
```

Some regular expressions are equivalent; choose the one that you prefer

Shorthand	Equivalent Range	Description
`\d`	`[0-9]`	Matches any digit (0-9).
`\D`	`[^0-9]`	Matches any non-digit character.
`\w`	`[a-zA-Z0-9_]`	Matches any word character (letters, numbers, and the underscore).
`\W`	`[^\w]`	Matches any non-word character (e.g., symbols, spaces).
`\s`	`[ \t\n\r\f\v]`	Matches any whitespace character (space, tab, newline, etc.).
`\S`	`[^\s]`	Matches any non-whitespace character.

Example - finding patterns in a VCF file

```
1   920760 rs80259304 T   C   .   PASS
AA=T;AC=18;AN=120;DP=190;GP=1:930897;BN=131   GT:DP:CB   0/1:1:SM
0/0:4:SM...
```

Find one record with at least one sample, that is, having genotype fields such as `0/0:1:SM` `0/0:4:SM`

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```

Find one record with at least one sample, that is, having genotype fields such as `0/0:1:SM` `0/0:4:SM`

- `"[01]/[01]"` (or `"\d/\d"`)
- `\s\d/\d:`

Example - finding patterns in a VCF file

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Find one record with at least one sample, that is, having genotype fields such as **`0/0:1:SM`** **`0/0:4:SM`**

- `"[01]/[01]"` (or `"\d/\d"`)
- `\s\d/\d:`

In [ ]:
```
!grep -E "[01]/[01]" ../downloads/genotypes_small.vcf | head -n 1
```

Example - finding patterns in vcf

```
1   920760 rs80259304 T   C   .   PASS
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```

- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

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```

- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

```
... 1/1:...  ... 1/1:...  ...
```

Example - finding patterns in vcf

`1  920760 rs80259304 T  C  .  PASS  AA=T;AC=18;AN=120;DP=190;GP=1:930897;BN=131  GT:DP:CB  0/1:1:SM  0/0:4/SM...`

- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

`... 1/1:...  ... 1/1:...  ...`

- `.*1/1.*1/1.*`
- `.*\s1/1:.*\s1/1:.*`

In [ ]:
```
!grep -E ".*\s1/1:.*\s1/1:.*" ../downloads/genotypes_small.vcf  | wc -l
```

Example from the project (with GTF file):

how many transcripts can the CFTR gene (ENSG00000001626) generate?

In [ ]:
```
!grep -E "transcript.*ENSG00000001626" ../assignment/data/Homo_sapiens.G
```

Cheat sheet

- `.` matches any character (once)
- `?` repeat previous pattern 0 or 1 times
- `*` repeat previous pattern 0 or more times
- `+` repeat previous pattern 1 or more times
- `\w` matches any letter or number, and the underscore
- `\d` matches any digit
- `\D` matches any non-digit
- `\s` matches any whitespace (spaces, tabs, ...)
- `\S` matches any non-whitespace
- `[abc]` matches a single character defined in this set {a, b, c}
- `[^abc]` matches a single character that is **not** a, b or c
- `[a-z]` matches any (lowercased) letter from the english alphabet
- `.*` matches anything, but **not cross multiple lines**.

Day 5, Exercise 1 (~30 min)

Practicing regular expressions

- Canvas -> Modules -> Day 5 -> Exercise 1 - day 5

Start the exercise by running

---

python retester.py

---

in the `downloads` folder in a terminal

Take a break after the exercise (~10 min)

Session 2

- How to use regex in Python
- Ex2: Regex using Python
- PyQuiz 5.1

Regular expressions in Python

Regular expressions in Python

In [ ]:

```
import re
pattern = r"col[ou]+r.*"
text = "The colour of the wall is very vibrant,\
but the color of the sky is even more spectacular."
# Try to find a hit
result = re.search(pattern, text)
if result:
    print(result.group())
    # print(result.span())
    # print(result.span()[0])
    # print(result.start())
else:
    print("No match found")
```

Regular expressions in Python

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but the color of the sky is even more spectacular."
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if result:
    print(result.group())
    # print(result.span())
    # print(result.span()[0])
    # print(result.start())
else:
    print("No match found")
```

`result.group()` : Return the string matched by the expression

`result.start()` : Return the starting position of the match

`result.end()` : Return the ending position of the match

`result.span()` : Return both (start, end)

Raw string

```
pattern = r"col[ou]+r.*"
```

- A raw string is a Python string literal prefixed with the letter `r`.
- Highly recommended to use raw string `(r"...")` for regular expression patterns in Python.
- Raw strings simplify pattern writing by preventing Python from interpreting backslashes `\` as special escape sequences.

Raw string

```
pattern = r"col[ou]+r.*"
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- Highly recommended to use raw string `(r"...")` for regular expression patterns in Python.
- Raw strings simplify pattern writing by preventing Python from interpreting backslashes `\` as special escape sequences.

In [ ]:
```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = "\bthe\b" # \b matches the word boundary, try with and without
                    # without 'r', \b is interpreted as ASCII backspace
result = re.search(pattern, text)
if result:
    print(result.group())
else:
    print("No match found")
```

In [ ]:
```

```

In [ ]:

```
# How if we only want to search for the words "colour", "color"?
pattern = r"col[ou]+r.*"
text = "The colour of the wall is very vibrant, but the color of the sky
# pattern = r"col[ou]+r\b"
result = re.search(pattern, text)
if result:
    print(f"Found text: '{result.group()}'")
    print(result.span())
else:
    print("No match found")
```

How to find all occurences of "color" variations?

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```
re.findall
```

How to find all occurences of "color" variations?

```
re.findall
```

In [ ]:
```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = r"col[ou]+r\b"
result = re.findall(pattern, text)
print(result)
```

How to find all occurences of "color" variations with their positions in the text?

How to find all occurences of "color" variations with their positions in the text?

```
re.finditer
```

How to find all occurences of "color" variations with their positions in the text?

```
re.finditer
```

In [ ]:

```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = r"col[ou]+r\b"

for result in re.finditer(pattern, text):
    print(f"Find word: {result.group()} at {result.span()}")
```

re.compile

re.compile

In [ ]:
```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = r"col[ou]+r\b"
p_find_colour = re.compile(pattern)
#print(type(p_find_colour))
p_find_colour.search(text)
# re.search(pattern, text)
# p_find_colour.findall("colur colour color")
```

re.compile

In [ ]:
```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = r"col[ou]+r\b"
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In [ ]:
```
p_find_colour.findall(text)
```

re.compile

In [ ]:
```
text = "The colour of the wall is very vibrant, but the color of the sky
pattern = r"col[ou]+r\b"
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#print(type(p_find_colour))
p_find_colour.search(text)
# re.search(pattern, text)
# p_find_colour.findall("colur colour color")
```

In [ ]:
```
p_find_colour.findall(text)
```

In [ ]:
```
for result in p_find_colour.finditer(text):
    print(f"Find colour at the position {result.span()}, the word is '{r
```

Benefits of using `re.compile`

- Improved performance - saves time by compiling the regex only once, especially when used in a loop
- Reusability - the compiled regex can be used multiple times
- Early error detection - syntax error of the pattern can be detected at the compiling stage instead of mixed with the other `re` functions

Case insensitiveness

In [ ]:
```
# Remember, [a-z]+ matches any lower case English word
pattern = r'[a-z]+'
result = re.search(pattern, '123 ATGAAA 456')
if result:
    print(result)
else:
    print("No match found")
```

Case insensitiveness

In [ ]:
```
# Remember, [a-z]+ matches any lower case English word
pattern = r'[a-z]+'
result = re.search(pattern, '123 ATGAAA 456')
if result:
    print(result)
else:
    print("No match found")
```

In [ ]:
```
pattern = r'[a-z]+'

result = re.search(pattern, '123 ATGAAA 456', flags=re.IGNORECASE)
if result:
    print(f"Found hit {result.group()} at the position {result.span()}")
else:
    print("No match found")
```

How to find a full stop?

In [ ]:
```
text = "The first full stop is here: ."

result = re.search(r".", text)
print(f"Found {result.group()} at position {result.start()}")
```

How to find a full stop?

In [ ]:
```
text = "The first full stop is here: ."

result = re.search(r".", text)
print(f"Found {result.group()} at position {result.start()}")
```

In [ ]:
```
# Use the escape character \
result = re.search(r"\.", text)
print(f"Found {result.group()} at position {result.start()}")
```

More operations

- `\` escaping a character, use the character as is
- `^` beginning of the string r"^abc"
- `$` end of string
- `|` boolean `or`

Matching beginning and end of line

```
^hello$
```

Matching beginning and end of line

`^hello$`

In [ ]:
```
text = "hello   \nhello\n    hello"
# print(text)
result = re.search(r'^hello$', text, flags=re.MULTILINE)
if result:
    print(f"Find word '{result.group()}' at position {result.span()}")
else:
    print("No match found")
```

Substitution

Substitution

In [ ]:
```
text = "Do it   becuase   I say so,     not becuase you want!"
# Spell the word because correctly
re.sub(r'becuase', 'because', text)
print(text)
```

Substitution

In [ ]:
```
text = "Do it   becuase   I say so,     not becuase you want!"
# Spell the word because correctly
re.sub(r'becuase', 'because', text)
print(text)
```

In [ ]:
```
text = re.sub(r'becuase', 'because', text)
print(text)
```

Substitution

In [ ]:
```
text = "Do it   becuase   I say so,     not becuase you want!"
# Spell the word because correctly
re.sub(r'becuase', 'because', text)
print(text)
```

In [ ]:
```
text = re.sub(r'becuase', 'because', text)
print(text)
```

In [ ]:
```
# Remove additional spaces
re.sub(r'\s+', ' ', text)
```

Example from Day 2 when lstrip() does not work as expected

In [ ]:
```
line = "AB:A:B should not be removed"
line.lstrip("AB:") # The chars argument is not a prefix
```

In [ ]:
```
line.removeprefix("AB:")
```

Example from Day 2 when lstrip() does not work as expected

In [ ]:
```
line = "AB:A:B should not be removed"
line.lstrip("AB:") # The chars argument is not a prefix
```

In [ ]:
```
line.removeprefix("AB:")
```

In [ ]:
```
newline = re.sub(r"^AB:", "", line)
print(newline)
```

Overview

- Construct regular expressions

```
p = re.compile()
```

- Searching

```
p.search(text)
re.search(pattern, text)
```

- Substitution

```
p.sub(replacement, text)
```

Typical code structure for text matching:

```
pattern = re.compile(r"regex")
result = pattern.search('string goes here')

if result:
    print('Match found: ', result.group())
else:
    print('No match')
```

Summary

- A powerful tool to search and modify text
- There is much more to read in the docs
- Note: regex comes in different flavours. If you use it outside Python, there might be small variations in the syntax.

Day 5, Exercise 2 (~30 min)

Use regular expressions with Python

- Canvas -> Modules -> Day 5 -> Exercise 2 - day 5
- Do bonus exercise only when you are super interested in RegEx

Take a break after the exercise (~10 min)

PyQuiz 5.1 (~10 min)

Lunch

Sum up!

Processing files - looping through the lines

```
with open('myfile.txt', 'r') as fh
    for line in fh:
        do_stuff(line)
```

In [ ]:

```
with open('myfile.txt', 'r', encoding='utf-8') as fh
    for line in fh:
        do_stuff(line)
```

Store values

```
iterations = 0
information = []

with open('myfile.txt', 'r') as fh:
    for line in fh:
        iterations += 1
        information += do_stuff(line)
```

Values

- Base types:

```
str     "hello"
int     5
float   5.2
bool    True
```

- Collections:

```
list  ["a", "b", "c"]
dict  {"a": "alligator", "b": "bear", "c": "cat"}
tuple ("this", "that")
set   {"drama", "sci-fi"}
```

Assign values

```
iterations = 0
score      = 5.2
# variable = literal
```

Compare and membership

```
+, -, *,...   # mathematical
and, or, not  # logical
==, !=        # (in)equality
<, >, <=, >=  # comparison
in            # membership
```

Showing the result: use `f-string`

In [ ]:
```
# Use f-string for print out result
value = 4
nextvalue = 1
nextvalue += value
print(f"nextvalue: {nextvalue}, value: {value}")
```

Be aware of the order of logical operators

In [ ]:

```
x = 5
y = 7
z = 0

x > 4 or y == 7  and z > 1
```

Be aware of the order of logical operators

In [ ]:
```
x = 5
y = 7
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x > 4 or y == 7  and z > 1
```

In [ ]:
```
(x > 4 or y == 7) and z > 1
```

Strings

Works like a list of characters

In [ ]:
```
mystr = "one"
```

In [ ]:
```
mystr += " two" # string concatnation
mystr
```

In [ ]:
```
len(mystr) # get the length
```

In [ ]:
```
"one" in mystr # in operator for substring matching
# mystr.find("one") # return index if find a hit
# mystr.index("one") # Identical to .find(), but handles failure differe
```

String is immutable

In [ ]:
```
mystr = "one"
mystr[1] = "W"
```

String is immutable

In [ ]:
```
mystr = "one"
mystr[1] = "W"
```

In [ ]:
```
mystr = "one"
print(mystr)
mystr = "two"
print(mystr)
```

String is immutable

In [ ]:
```
mystr = "one"
mystr[1] = "W"
```

In [ ]:
```
mystr = "one"
print(mystr)
mystr = "two"
print(mystr)
```

In [ ]:
```
mystr = "one"
print(f"mystr = {mystr}, address = {id(mystr)}")
mystr = "two"
print(f"mystr = {mystr}, address = {id(mystr)}")
```

String manipulation

```
s.strip()  # remove trailing white spaces

s.split()  # split the line into columns

s.upper(), s.lower()  # change the case
```

Regular expressions help you find and replace strings

```
p = re.compile(r'A.A.A')
p.search(dna_seq)

p = re.compile(r'T')
rna_seq = p.sub('U', dna_seq)
```

Regular expressions help you find and replace strings

```
p = re.compile(r'A.A.A')
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```

In [ ]:

```
import re
# the greedy star (*)! Match as long as possible
re.search(r'p.*\sp', 'a python programmer writes python code').group()
```

Regular expressions help you find and replace strings

```
p = re.compile(r'A.A.A')
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In [ ]:
```
import re
# the greedy star (*)! Match as long as possible
re.search(r'p.*\sp', 'a python programmer writes python code').group()
```

In [ ]:
```
# To match the shortest possible string, use non-greedy quantifier *?
re.search(r'p.*?\sp', 'a python programmer writes python code').group()
```

Collections

Can contain strings, integer, booleans...

- Most collections are mutable (except **tuple**): you can *add*, *remove*, *change* values
- Lists:

```
mylist.append('value')
```

- Dicts:

```
mydict['key'] = 'value'
```

- Sets:

```
myset.add('value')
```

Collections

- Test for membership:

```
value in myobj
```

- Check size:

```
len(myobj)
```

Lists

- Ordered!

```
mylist = [1, 8, 1, 2, 4]

mylist.sort()
mylist.reverse()
mylist[2]
mylist[-1]
mylist[2:6]
```

In [ ]:
```
mylist = [1, 8, 1, 2, 4]
mylist
```

In [ ]:
```
mylist.sort()
print(mylist)
```

In [ ]:
```
mylist.reverse()
print(mylist)
```

In [ ]:
```
mylist[2]
```

In [ ]:
```
mylist[-1]
```

In [ ]:
```
mylist[2:]
```

Dictionaries

- List of key value pairs

```
mydict = {"a": "alligator", "b": "bear", "c": "cat"}
counter = {"cats": 55, "dogs": 8}

mydict["a"]
mydict.keys()
mydict.values()
mydict.items() # get both keys and values
```

In [ ]:

```
counter = {'cats': 0, 'others': 0}
# counter = {}
for animal in ['zebra', 'cat', 'dog', 'cat']:
    if animal == 'cat':
        counter['cats'] += 1
    else:
        counter['others'] += 1

print(counter)
```

Sets

- Bag of values
    - No order
    - No duplicates
    - Fast membership checks
    - Logical set operations (union, difference, intersection...)

```
myset = {"drama", "sci-fi"}

myset.add("comedy")

myset.remove("drama")
```

In [ ]:

```
set1 = set(["1", "2", "3", "4", "5"])
set1
```

In [ ]:
```
set1 = set(["1", "2", "3", "4", "5"])
set1
```

In [ ]:
```
set1.add("1")
set1
```

In [ ]:
```
set1 = set(["1", "2", "3", "4", "5"])
set1
```

In [ ]:
```
set1.add("1")
set1
```

In [ ]:
```
set2 = set(["3", "6"])
set1.intersection(set2)
```

In [ ]:
```
set1 = set(["1", "2", "3", "4", "5"])
set1
```

In [ ]:
```
set1.add("1")
set1
```

In [ ]:
```
set2 = set(["3", "6"])
set1.intersection(set2)
```

In [ ]:
```
set1.union(set2)
```

In [ ]:
```
set1 = set(["1", "2", "3", "4", "5"])
set1
```

In [ ]:
```
set1.add("1")
set1
```

In [ ]:
```
set2 = set(["3", "6"])
set1.intersection(set2)
```

In [ ]:
```
set1.union(set2)
```

In [ ]:
```
set1.difference(set2)
```

Tuples

- A group (usually two) of values that belong together

```
tup = (max_length, sequence)
```

- An ordered sequence (like lists)

```
length = tup[0]  # get content at index 0
```

- Immutable

Tuples

- A group (usually two) of values that belong together

```
tup = (max_length, sequence)
```

- An ordered sequence (like lists)

```
length = tup[0]  # get content at index 0
```

- Immutable

In [ ]:
```
tup = (2, 'xy')
tup[0]
# type(tuple((list(tup)))) # tuple and list can be converted
```

In [ ]:
```
tup[0] = 2
```

Tuples in functions

```
def find_longest_seq(file):
    # some code here...
    return length, sequence
```

In [ ]:

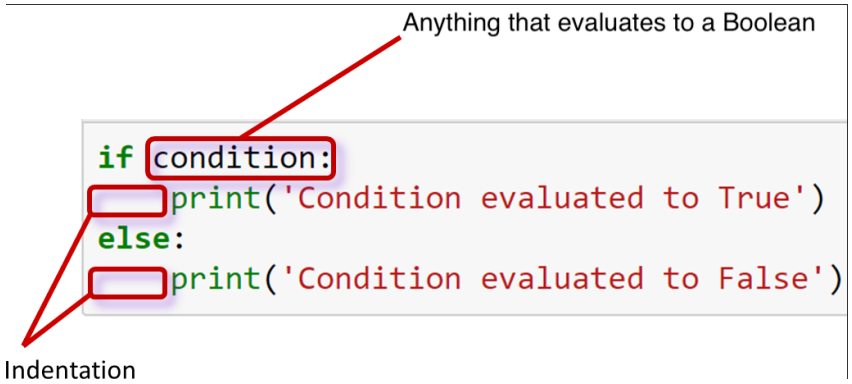
```
def find_longest_seq(file):
    length = 10
    sequence = "AAAAATTTTT"
    return length, sequence

answer = find_longest_seq("dummyfile") # return as a tuple
print(type(answer))
# _, sequence = find_longest_seq("dummyfile") # return as two variables
# print(sequence)
```

Deciding what to do with if else statement

In [ ]:
```
count = 8
if count > 10:
    print('big')
elif count > 5:
    print('medium')
else:
    print('small')
```

Deciding what to do - if statement



`if x:` is equvalent to `if bool(x):`

In [ ]:
```
if [1,4,6]:
    print("yes")
```

Program flow - for loops

In [ ]:
```
information = []
with open('myfile.txt', 'r') as fh
    for line in fh:
        if is_comment(line):
            use_comment(line)
        else:
            information.append(read_data(line)) # read_data return a lis
```

Program flow - while loops

In [ ]:
```
information = []
with open('myfile.txt', 'r') as fh:
    # Read the first line
    line = fh.readline()

    # Continue to read lines until an empty string is returned
    while line:
        information.append(read_data(line)) # read_data return a list
        line = file.readline()  # Read the next line
```

Different types of loops

`For` loop

is a control flow statement that performs operations over a known amount of steps.

`While` loop

is a control flow statement that allows code to be executed repeatedly based on a given Boolean condition.

Which one to use?

`For` loops - standard for iterations over lists and other iterable objects

`While` loops - more flexible and can iterate an unspecified number of times

In [ ]:

```
# For loop example
# You know the number of iterations before hand
user_input = "thank god it's friday"
for letter in user_input:
    print(letter.upper())
```

In [ ]:
```python
# For loop example
# You know the number of iterations before hand
user_input = "thank god it's friday"
for letter in user_input:
    print(letter.upper())
```

In [ ]:
```python
# While loop example
# The number of iterations is unknown before hand
key = ""
max_try = 3
cnt = 0
TARGET_KEY = "python2026"
while key != TARGET_KEY:
    key = input("Enter your key: ")
    cnt += 1
    if cnt > max_try:
        print("WARNING!! Too many failures!")
        break
    if key == TARGET_KEY:
        print("Correct key!")
```

Controlling loops

- `break` - stop the loop
- `continue` - go on to the next iteration

In [ ]:

```
# example for break
user_input = "thank god it's friday"
for letter in user_input:
    if letter == 'd':
        break
    print(letter.upper())
```

In [ ]:
```python
# example for break
user_input = "thank god it's friday"
for letter in user_input:
    if letter == 'd':
        break
    print(letter.upper())
```

In [ ]:
```python
# example for continue
user_input = "thank god it's friday"
for letter in user_input:
    if letter == ' ' or letter == '\'':  # Skip spaces and apostrophes
        continue
    print(letter.upper())
```

Watch out!

In [ ]:
```
# DON'T RUN THIS
i = 0
-while i < 10:
    print(user_input[i])
```

Watch out!

In [ ]:
```
# DON'T RUN THIS
i = 0
-while i < 10:
    print(user_input[i])
```

While loops may be infinite!

File Input/Output

- In: Read from files

```
with open(filename, 'r') as fh:
    for line in fh:
        do_stuff(line)
```

  Read information from command line: `sys.argv[1:]`

- Out: Write to files:

```
with open(filename, 'w') as fh:
    fh.write(text)
```

- Printing:

```
print('my_information')
```

Input/Output

Open files should be closed:

```
      fh.close()
```

or use the `with` clause

```
    with open(filename, "r") as fh:
        do_something
```

Code structure

- Functions (unit)
- Modules (the file)
- Packages (the folder)
- Libraries (the large collection)

Functions

- A named piece of code that performs a certain task.

```
def functionName(arg1, arg2, arg3):

    finalValue = 0

    # Here is some code where you can do
    # calculations etc, on arg1, arg2, arg3
    # and update finalValue

    return FinalValue
```

- Is given a number of input arguments
    - to be used within the function body
- Returns a result (maybe `None`)

Functions - arguments

```
def prettyprinter(name, value, delim=":", end=None):
    out = "The " + name + " is " + delim + " " + value
    if end:
        out += end
    return out
```

- used to set default values (often `None`)
- can be skipped in function calls, then the arguments are positional
- improve readability when keys are used

Using your code

Any longer pieces of code that have been used and will be re-used should be saved

- Save it as a file `mycode.py`
- To run it: `python3 mycode.py` or `python mycode.py`
- Import it: `import mycode`

Documentation and comments

```
""" This is a doc-string explaining what the purpose of this function/
module is """
```

```
# This is a comment that helps understanding the code
```

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```
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module is """
```

```
# This is a comment that helps understanding the code
```

- Comments *will* help you
- Undocumented code rarely gets used
- Try to keep your code readable: use informative variable and function names

Why programming?

Endless possibilities!

- reverse complement DNA
- custom filtering of VCF files
- plotting of results
- dealing with excel files

Why programming?

- Computers are fast
- Computers don't get bored
- Computers don't get sloppy

Why programming?

- Computers are fast
- Computers don't get bored
- Computers don't get sloppy

- Create reproducable results
    - for you and for others to use
- Extract large amount of information

Type hinting: popular for modern Python coding

In [ ]:

```python
age: int = 30
name: str = "Bob"

def show_new_age(age: int, name: str) -> int:
    # 'age' is expected to be an integer
    # 'name' is expected to be a string
    # The function is expected to return an integer
    new_age: int = age + 10
    print(f"New age of {name} is {new_age}")
    return new_age

show_new_age(age, name)
```

Type hinting: popular for modern Python coding

In [ ]:
```python
age: int = 30
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def show_new_age(age: int, name: str) -> int:
    # 'age' is expected to be an integer
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    # The function is expected to return an integer
    new_age: int = age + 10
    print(f"New age of {name} is {new_age}")
    return new_age

show_new_age(age, name)
```

- Type hinting improve the documentation and readability
- With proper tools and IDE, type hinting helps early detection of errors and debugging

In [ ]:

```
age: int = 30.0
name: str = "Bob"
print(type(age))
```

In [ ]:
```
age: int = 30.0
name: str = "Bob"
print(type(age))
```

Type hinting does not change how Python interprets your code

Why learning programming

- AI agents are already very powerful and keep evolving rapidly
- AI still makes mistakes
- In research, the answer or expected result format is usually unknown beforehand

Why learning programming

- AI agents are already very powerful and keep evolving rapidly
- AI still makes mistakes
- In research, the answer or expected result format is usually unknown beforehand

- You know the basics - don't be afraid to try, it's the only way to learn
- You will get faster

Final advice

- Getting help
    - Ask colleauges
    - Try speak out your problem
    - Search the web
    - Ask AI and use AI (e.g. Claude, Gemini, ChatGPT, Muse), but with caution.
    - Don't spend too much time on memorizing function and library names etc unless it's your hobby
    - NBIS drop-ins

Now you know Python!



Well done!

You will get a certificate after the course

NBIS presentation

Google Form for anonymous evaluation

Just a small exercise to finish the day and have fun!

Canvas -> Module -> Day 5 -> Exercise 3 - day 5

Exercise 3

In [ ]:
```
!cp ../downloads/checker.py .
# !copy ../downloads/checker.py . # on Windows probabaly
```

In [ ]:
```
import checker
```

In [ ]:
```
checker.check()
```

In [ ]:
```
name = "Shu"
checker.check_name(name)
```

In [ ]:
```

```